Week 14

Spring 2027 · Section 017 · MW 7:30–8:50 PM · RRC Room 8210

Course Level Objectives:

- Explain the use of data collection and statistics as tools to reach reasonable conclusions. (CLO 1)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Classify a study design as paired or independent samples. (CLO 1, CLO 8) **2.** Define the difference variable d and the order of subtraction. (CLO 8) **3.** Perform a paired t test. (CLO 8) **4.** Construct and interpret a paired t confidence interval. (CLO 6) **5.** Explain why pairing reduces variability. (CLO 8)	e-Text: 6.5 Paired Difference in Means Interactive Lecture: 6.5 Paired Difference in Means Slides: 6.5 Paired Difference in Means	Participation: Paired or Independent? – define d and the order of subtraction Your Turn: Paired or Unpaired? (class discussion of study scenarios)	Online Homework: 6.5 Paired Difference in Means (due Sun May 2, 11:59 PM) Take-Home Quiz 9 (due Wed Apr 28, 11:59 PM) – covers 6.2, 6.4, 6.5 (all objectives) Test 3 (Fri Apr 30 – Sun May 2, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Apr 29) (all objectives)	WileyPLUS StatKey Blackboard Respondus
6. State the hypotheses for a chi-square test of association from a two-way table. (CLO 8) **7.** Compute the expected counts for a two-way table. (CLO 8) **8.** Check that all expected counts are at least 5. (CLO 8) **9.** Compute the chi-square statistic and its degrees of freedom, $(r-1)(c-1)$. (CLO 8) **10.** Interpret the right-tail p-value of a chi-square test. (CLO 8) **11.** Relate the chi-square test on a 2×2 table to the two-proportion z test. (CLO 8) **12.** Explain why a significant result shows association but not direction or causation. (CLO 8)	e-Text: 7.2 Testing for an Association Between Two Categorical Variables Interactive Lecture: 7.2 Testing for an Association Between Two Categorical Variables Slides: 7.2 Testing for an Association Between Two Categorical Variables StatKey – Theoretical Distributions (χ^2)	Participation: Chi-Square Test Your Turn: Faculty Rank and Gender: compute expected counts, cell contributions, χ^2 and df; decide and interpret	Online Homework: 7.2 Testing for an Association Between Two Categorical Variables (due Sun May 2, 11:59 PM) Take-Home Quiz 10 (due Mon May 10, 11:59 PM) – covers 7.2, 8.1, 9.1 (all objectives)	WileyPLUS StatKey Blackboard

Also this week: Last day to withdraw – Mon Apr 26 (before Test 3)